SMART INDIA HACKATHON 2026

Smart Intelligent Response and Emergency Navigation(S.I.R.E.N)

Problem Statement ID - SIH26222

Problem Statement Title - Student Innovation: ideas to address the growing pressures on the city's resources, transport networks, and logistic infrastructure.

Theme - Transportation & Logistics

PS Category - Hardware

Team ID - team-h16

Team Name (Registered on portal) - TVARIT

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S.I.R.E.N

SMART INTELLIGENT RESPONSE AND EMERGENCY NAVIGATION

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An IoT + AI framework to clear the actual path — not just the traffic signal.

⚠ PROBLEM

In India, ambulances face critical **delays** because the problem extends far **beyond traffic signals.** Current smart systems are completely blind to the actual **physical blockages** on the road.

The Local Lane Trap:
Ambulances frequently get **trapped** on semi-urban roads or local lanes that have **no traffic lights** at all.

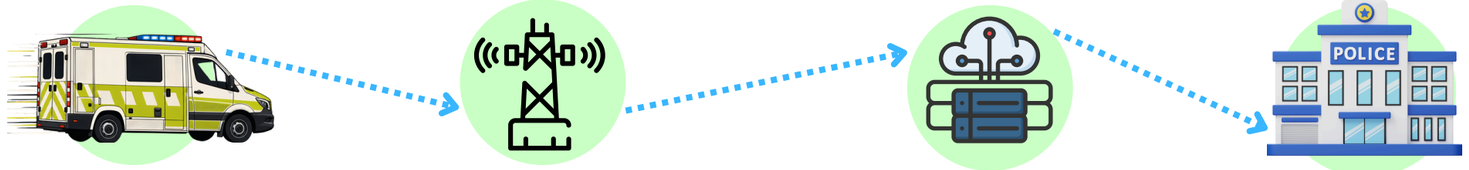
Mid-Block Obstructions:
Emergency vehicles are blocked by unpredictable factors like **unruly crowds**, roadside **disputes**, or vendors.

The Fatal Blind Spot:
Ambulances remain **stuck 200–500 meters** away from major junctions because existing systems falsely assume a green light means a clear path.

Solution

A layered IoT + AI system that manages the **ambulance's entire route**—not just traffic signals. It fuses acoustic detection, **computer vision and V2X/cloud coordination** to detect and decide all obstructions causing delay. The system degrades gracefully, every layer still works if other fails.

RFID **T-ERS (Kerala) — RFID + GPS [1]** **Signal pre-emption** 300–500 m from junction	**Problem:** Needs tagged hardware on every vehicle; **blind to mid-block jams** and non-signal roads **Ours: RoIP** and **computer vision** solves it.
B-ATCS (Bengaluru) AI vision + CoSiCoSt [2] City-wide **green wave** via cellular GPS	**Problem:** Depends on 4G/5G; **fails in urban canyons**; no local-lane or crowd-blockage awareness. **Ours:LoRa** and **edge nodes** work as back up if network fails.
Green Corridors (Manual) Near-perfect priority on **planned routes**	**Problem:** Causes **severe secondary gridlock** across the city. **Ours:** Uses **Reinforcement Learning (RL)** to instantly assign compensatory green time to cross-traffic.
Nagpur IITMS [3] Centralised ANPR + **adaptive signal** design	**Problem:** Rollout/vendor failure; even on paper, doesn't address **in-lane obstructions**. **Ours: RoIP** addresses in-lane obstructions.



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TECHNICAL-APPROACH

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1 Edge Devices (The Eyes & Ears)

Continuous Streaming: Ambulances constantly transmit real-time **GPS** telemetry and **SOS signals**.
Hardware Failsafe: If connectivity drops, intersection **CCTV** cameras and **microphones** detect the approaching ambulance via sight or siren sound.
Preemptive Action: The nodes automatically turn lights green and trigger a **10-second** public **announcement** to safely clear the junction.

2 Real-Time Data Hub (The Nervous System)

High-Speed Ingestion: Uses an **MQTT** broker to bypass slow web requests, capturing thousands of live location updates per second.
Traffic Buffering: A **Redis** buffer temporarily holds fast-moving data so the backend never gets overwhelmed during large-scale fleet operations.

3 Cloud Decision Engine (The Brain)

Vector-Based Preemption: Uses geospatial vector math to process live streams, calculating the exact trajectory and heading to accurately **turn lights green** strictly along the active path.
Gridlock Prevention: A dedicated **ML** model calculates the "**time loss**" inflicted on civilian traffic while the light is held green.
Dynamic Rebalancing: Intelligently adjusts subsequent **red/green cycles** to quickly flush out congestion and prevent city-wide traffic jams.

4 Dispatch & Dashboards (The Control Room)

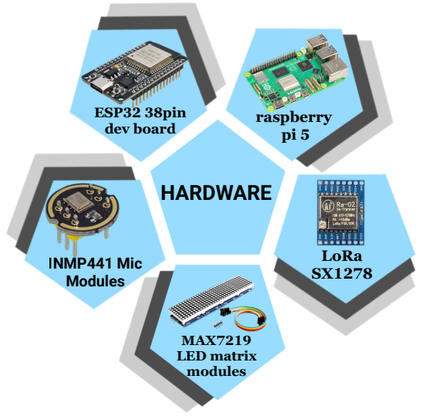
Interactive Interface: Provides a live, **WebSocket-powered map** for emergency dispatchers.
Real-Time Tracking: Allows human operators to **monitor** emergency routes, fleet locations, and congested intersections seamlessly without page refreshes.

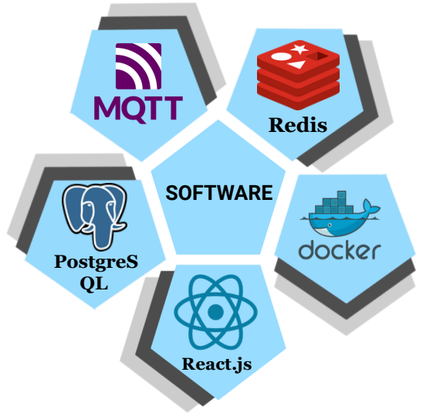
5 Unified Alerting Gateway (The Radio Network)

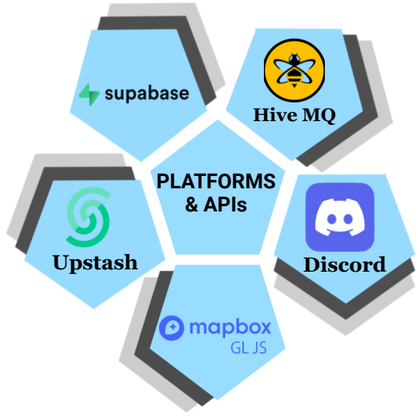
Targeted Dispatch: Generates Text-to-Speech audio and pushes it specifically to the nearest police units stationed along the active emergency path using **Radio over IP (RoIP).**
Legal Simulation: While real-world deployment requires restricted VHF/UHF **police frequencies (136–174 MHz)**, Discord voice channels are used to perfectly and legally simulate this hardware for the demo.

6 Geospatial Data Lake (The Memory)

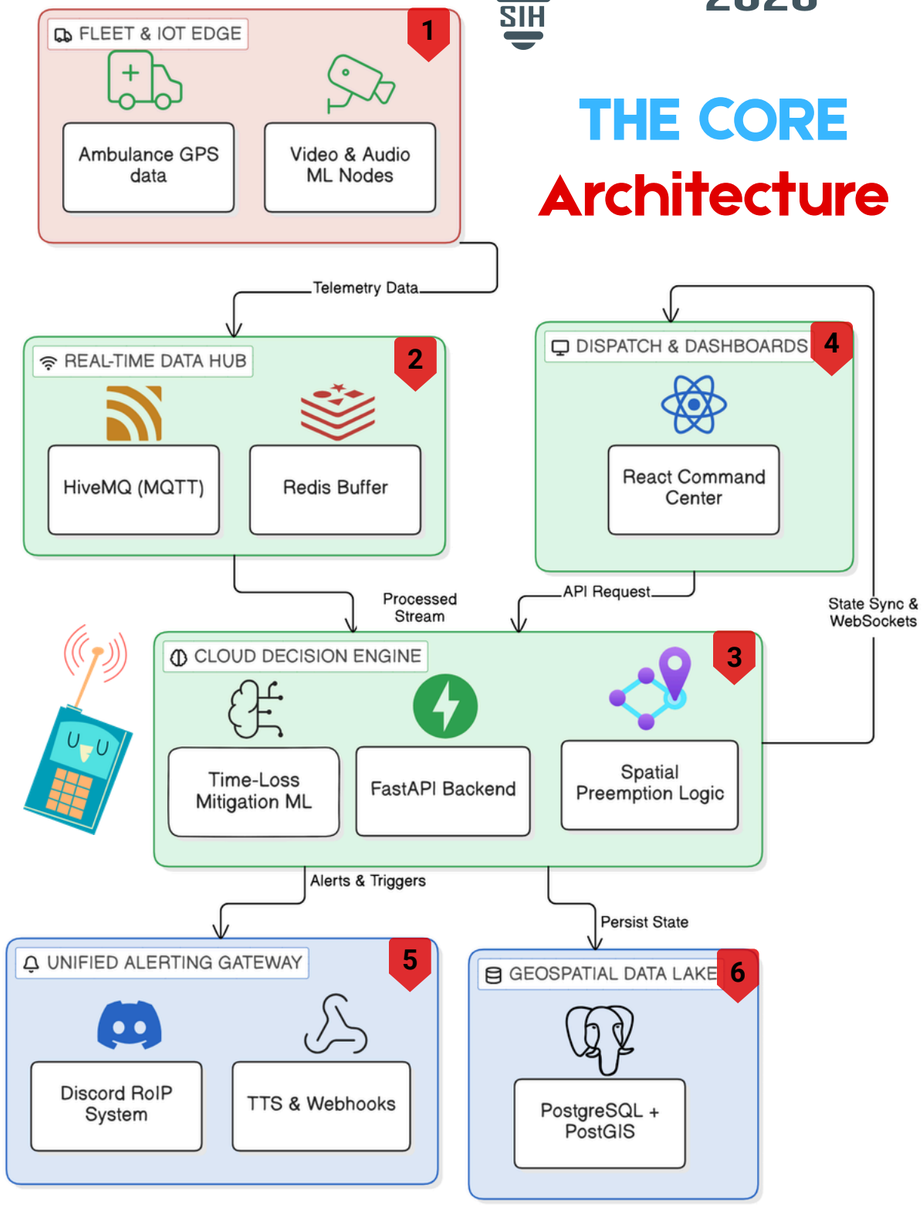
City Mapping: Permanently stores the physical layout of the city, including hospitals, traffic signals, and road networks.
Complex Math: Leverages the **PostGIS** extension to query exact geographic coordinates and shapes to power the system's advanced **spatial routing** algorithms.







THE CORE Architecture



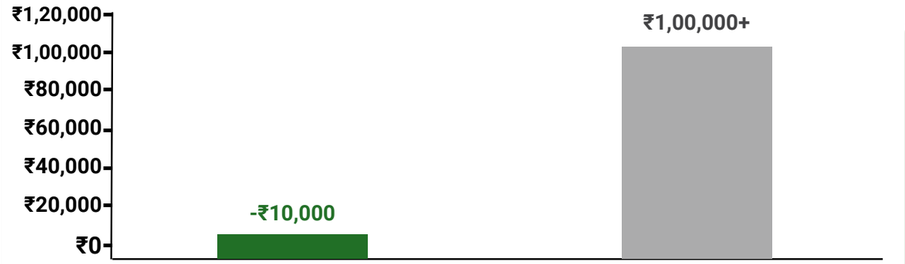
FEASIBILITY AND VIABILITY

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FEASIBILITY

COST COMPARISON PER INTERSECTION UPGRADE (₹)



Up to 90% more cost-effective per intersection upgrade

Ultra-Low Cost Infrastructure

Ambulance IoT Units:
Basic GPS and telemetry tracking costs just **₹2,500–₹3,800** per vehicle.

Direct RF Integration:
An additional ₹300–₹500 adds **VHF/UHF** transceivers, broadcasting automated **SOS routing** directly to existing police **walkie-talkies** without new radio towers.

Smart Retrofitting:
Leverages existing traffic **CCTV poles**. We only add low-cost acoustic microphones (for siren detection) and solar panels, ensuring grid-independent, **zero-cost** power.

Seamless Legacy Integration

Plug-and-Play Hardware:

Requires **minute architectural changes.** IoT relays patch directly into legacy traffic switchboards without requiring physical road restructuring.

Dynamic "Virtual Lanes":

Solves the spatial impossibility of building dedicated **"Yellow Ambulance Lanes"** in congested Indian cities by preemptively clearing standard traffic lanes via software.

VIABILITY

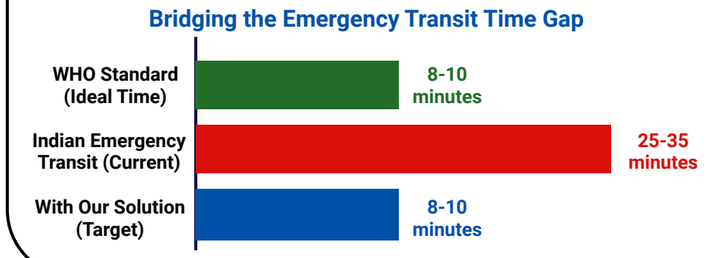
Reclaiming the "Golden Hour"

The "Transport-Only" Reality:
Many Indian ambulances function purely as transport vehicles **lacking basic medical facilities,** making **rapid transit** to a hospital the absolute sole factor for survival.

Bridging the Standard Gap:
Indian emergency transit frequently takes **25–35** minutes, severely failing the **WHO's 8–10** minute standard.

Maximized Survival Rate: Treating trauma within the 60-minute **"Golden Hour"** reduces mortality by **50%**. Eradicating intersection wait times mathematically reclaims this window, targeting the **1.73 lakh** lives lost annually on Indian roads.

THE IMPACT: RECLAIMING PRECIOUS MINUTES



Reclaim the Golden Hour can reduce mortality by 50% and help save 1.73 lakh lives annually on Indian roads.

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IMPACT AND BENEFITS

Building Faster Responses. Saving More Lives.

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The Impact

Life-Saving & Systemic Outcomes

1. MASSIVE-SCALE LIFE SAVING

India loses over 1.73 lakh lives annually to road accidents, with **up to 30% of fatalities linked to delayed ambulances**. Guaranteeing a green corridor mathematically saves **tens of thousands of lives** currently lost to simple traffic grids.

2. RECLAIMING THE GOLDEN HOUR

Every 1-minute reduction in emergency response improves severe trauma survival by **7% to 10%**. We slash the Indian urban average from **35+ minutes** down to **under 15 minutes**, rapidly approaching the **WHO's 8-minute standard**.

3. THE "TRANSPORT-ONLY" REALITY

Over 70% of Indian ambulances **lack Advanced Life Support** (ALS). Because critical care can begin in transit, eliminating the **30% to 40%** of time wasted at red lights acts as the absolute determinant for patient survival.

4. SOCIAL EQUITY

Eliminates the geographic penalty of congested urban zones. By neutralizing bottlenecks, rapid life-saving response becomes a **universal standard** for all citizens, rather than a privilege dictated by their location.

The Benefits

Operational & Governance Advantages

1. GOVERNMENT & SMART CITY SCALING (NEW)

Delivers civic authorities a **highly visible, life-saving** infrastructure upgrade without massive civil engineering budgets. It reforits seamlessly into existing traffic grids and generates vital **geospatial data** to help city planners optimize future road networks.

2. ERADICATING SECONDARY TRAUMA

Emergency vehicles forcing their way through active red lights face a **4.8x higher risk of severe collisions**. Preemptive green lights remove driver cognitive load, actively protecting medical crews and civilians from high-speed secondary crashes.

3. PINPOINT TACTICAL DISPATCHING

Replaces chaotic, city-wide radio chatter. The automated RoF system isolates the live emergency trajectory, alerting only the **2 to 3 specific traffic police** units needed to clear physical bottlenecks without disrupting the broader city police grid.

RESEARCH & REFERENCES

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REFERENCES

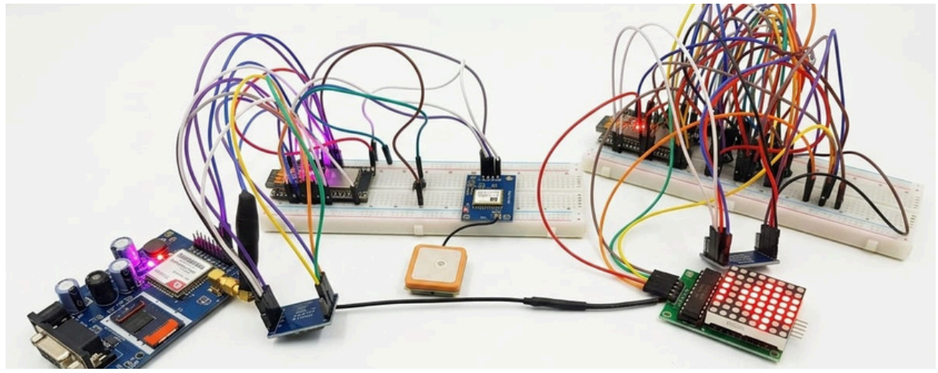
[1] Press Trust of India, “Kerala startup finds solution to end problem of ambulances stuck in traffic,” The Indian Express, Jun. 26, 2017. [AVAILABLE]

[2] Bengaluru Traffic Police, Government of Karnataka, “Adaptive Traffic Control System (ATCS),” Bengaluru, Karnataka, India. [AVAILABLE]

[3] Nagpur Updates Desk, “Nagpur Gets AI-Powered Traffic Control: 32 Junctions Live on IITMS, Work Underway at 80 More — ₹198 Crore Upgrade in Full Swing,” Nagpur Updates, May 11, 2026. [AVAILABLE].







OUR RESEARCH FOUNDATION

1 The Medical Crisis

1 in 5 ambulance deaths in India are caused directly by traffic delays. Survival rates **drop up to 12%** for every single minute of delay.

2 Acoustic Validation

Utilizing proven Time-Difference-of-Arrival **(TDOA) algorithms** and **CNN classifiers** for urban siren detection.

3 V2X Standards

Leveraging **600 MHz UHF** bands (ARIB STD T109) for communication, proven to **penetrate** urban concrete better than standard 4G/5G.

4 Algorithm Design

Built upon indigenous traffic logic like C-DAC's CoSiCoSt for **non-lane-based** Indian traffic.

KEY LINKS

[LINK] [LINK] [LINK]

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THE PEOPLE BEHIND SIREN

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